CITY UNIVERSITY OF HONG KONG

SUPPORTING DOCUMENTS

Academic Ranking and Recommendation Letters

Applicant Yongyi Lai

Programme MSc Computer Science (P53)

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深圳技术大学
SHENZHEN TECHNOLOGY UNIVERSITY

专业排名
RANKING CERTIFICATE

学生赖咏怡，女，学号202300202024，2005年04月03日生，于2023年09月考入我校，现在是计算机科学与技术专业四年制普通全日制（本科）学生，该生在校期间课程学习成绩专业综合排名在该专业156名学生中排名第6。

特此证明。

It is hereby certified that Lai Yongyi, female, Student ID No. 202300202024, born on 03 April 2005, was enrolled at Shenzhen Technology University in September 2023, majoring in Computer Science and Technology and ranking the 6 out of 156.

深圳技术大学教务部
Department of Academic Affairs
2026-01-18

地址：广东省深圳市坪山区兰田路3002号 邮编：518118 电话：+（86）0755-23256250 电子邮箱：jwb@sztu.edu.cn
Address : 3002 Lantian Road, Pingshan District, Shenzhen, Guangdong, China, 518118 Tel : +86 755 2325 6250 Email : jwb@sztu.edu.cn

深圳技术大学
SHENZHEN TECHNOLOGY UNIVERSITY

Recommendation Letter — Academic Advisor

Date: 21 September 2026

Dear Members of the Admissions Committee,

I am writing to support Yongyi Lai's application to the MSc in Computer Science at City University of Hong Kong.

I am Associate Dean in the School of Artificial Intelligence at Shenzhen Technology University, where Yongyi is a final-year student in Computer Science and Technology. I have known Yongyi since 2023, first as the instructor of University Computer Science A, and since then as the supervisor of two robotics projects in my laboratory. I have also followed Yongyi's competition and independent work closely, and this letter draws on all three.

Yongyi first came to my attention as a strong student who deliberately extended a Computer Science background into robotics. The academic record itself is straightforward: an average of 87.66 and a rank of 6th of 156 in the cohort. What stood out was that Yongyi had taken Principles of Automatic Control and Modern Control Theory in addition to the major curriculum, and then applied that preparation to real robot projects. This combination of computing ability and purposeful study in control theory was the basis on which Yongyi took on the two projects I describe below.

The first project, a wheel-legged robot with variable leg length, tested this preparation under changing mechanical geometry: the balance controller had to remain effective as the robot's leg configuration changed. Yongyi constructed the body-pitch and motion states from IMU, wheel-speed and joint feedback, implemented an LQR controller with gains scheduled by leg length, and added a virtual model controller that derived virtual leg length and posture from the closed-chain leg kinematics and mapped support force and pitch torque to the joints through the Jacobian. The controller was integrated and tuned on the physical robot, running on an STM32H7/FreeRTOS platform with FDCAN. The robot kept its body balanced and tracked velocity commands while its legs extended and retracted. The tuning process mattered to me as much as the result: when balance failed, Yongyi located the causes in state inconsistency, torque limits and ground-contact transitions and corrected each in turn, demonstrating a disciplined approach to diagnosing a physical system.

The second project expanded this work from locomotion to navigation. On a car-like mobile robot, Yongyi was responsible for the autonomy chain: EKF/AMCL localisation fusing IMU, wheel encoders and 2D LiDAR, including the handling of timing and coordinate-frame inconsistencies that cause pose

jumps; Hybrid-A* global planning under the vehicle's kinematic constraints; TEB local trajectory optimisation; and an MPC tracking controller on a car-like kinematic model. The system ran on the physical vehicle and achieved autonomous navigation from localisation through obstacle-aware planning to trajectory tracking. In my experience, it is unusual at undergraduate level for a student to take responsibility for work spanning localisation, planning and control and carry it through to a functioning vehicle.

I have also seen the same capabilities in work I did not supervise directly but followed closely. In the university RoboMaster team Yongyi was responsible for the embedded control of the hero robot, implementing the gimbal, launcher and chassis control loops on STM32F407/FreeRTOS over CAN; the robot competed in the 2025 national final, where the team won third prize, and the team also took second prize in the 3V3 league. Yongyi also designed and built, alone, an OpenHarmony smart gardening system and demonstrated it to me: it runs YOLOv5 inference on an RK3588 edge device, makes decisions through a LangGraph agent chain and controls irrigation on a Hi3861 device, a complete loop from perception to actuation.

Taken together, these four pieces of work show a student who can move from control theory to embedded implementation and back, who finishes systems rather than components, and who treats documentation and hardware safety as part of the design. They also show where project work stops. In both laboratory projects Yongyi arrived at the model by working it out for that particular robot, and at the gains by tuning on the machine; the general theory of control, planning and learning that would let the same student design rather than rediscover is not something our undergraduate curriculum provides. Yongyi recognises this and has chosen a taught master's for that reason, which I consider the right decision at the right time.

I recommend Yongyi Lai for admission without hesitation, and I would be glad to provide any further information.

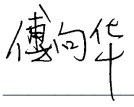
Yours sincerely,

Fu Xianghua

Associate Dean, School of Artificial Intelligence

Shenzhen Technology University

fuxianghua@sztu.edu.cn

Referee signature: 

SHENZHEN INSTITUTE
of ARTIFICIAL INTELLIGENCE AND ROBOTICS for SOCIETY
深圳市人工智能与机器人研究院

Recommendation Letter — AIRS Mentor

Date: 21 September 2026

Dear Members of the Admissions Committee,

I am writing to recommend Yongyi Lai for admission to the MSc in Computer Science at City University of Hong Kong.

I am a Robot Development Engineer at the Shenzhen Institute of Artificial Intelligence and Robotics for Society (AIRS), where I supervised Yongyi directly as a robotics software intern from September 2025 to March 2026. During those six months Yongyi was responsible for the chassis balance control of a three-omni-wheel ballbot carrying a dual-arm upper body, a demanding control problem in our group. Under my supervision, Yongyi made steady progress toward this objective.

The task was to keep a dynamically unstable base balanced while two arms moved on top of it, which shifts the centre of mass continuously. Yongyi built the control state from IMU and wheel feedback, designed a cascaded controller in which a position outer loop generates bounded attitude references for an attitude inner loop, allocated the resulting actions to the three wheels through the wheel kinematics, and deployed and tuned the whole loop on the physical robot. To handle the arm disturbance, Yongyi estimated the centre-of-mass offset from the arm joint states using the robot's URDF model with Pinocchio and fed it into the balance controller as gravity feedforward, then verified the effect by comparing the balance response with compensation switched on and off under fixed arm trajectories. By the end of the internship the chassis maintained dynamic balance while both arms executed their trajectories, which was the objective we had set.

What stood out was Yongyi's ability to return to model-based analysis when empirical tuning could no longer fully explain the robot's behaviour. Rather than continuing to adjust gains without a model, Yongyi derived a four-state inverted-pendulum model of the ballbot, linearised it and used LQR to analyse how the weights on position, velocity, tilt and tilt rate trade stability against control effort. That analysis became our reference for interpreting the deployed controller's gains, identifying state anomalies and locating actuator saturation. It also demonstrated analytical ability beyond what I would normally expect from an undergraduate intern.

Yongyi was equally reliable on the engineering safeguards that make a hardware loop trustworthy: ROS state processing and CAN motor execution, state initialisation on re-enable, output limiting, and fall detection and recovery. These safeguards were implemented carefully and documented clearly, allowing the robot to be tested repeatedly under controlled conditions. Yongyi communicated progress and

problems concisely, asked precise questions, and worked independently once a direction was agreed.

Yongyi discussed with me the plan to take a taught master's in robotics or AI before working as an algorithm engineer. Given how much of this internship's progress came from Yongyi turning to model-based analysis when tuning ran out, I think that is the right next step. On the evidence of this internship, Yongyi has the technical ability, the hands-on discipline and the intellectual curiosity to benefit fully from your programme. I recommend Yongyi without reservation and would be glad to answer any questions.

Sincerely,

Chen Dansheng

Robot Development Engineer

Shenzhen Institute of Artificial Intelligence and Robotics for Society (AIRS)

chendansheng@cuhk.edu.cn

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